Assignment 5

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Question 2-3

Write a program that takes the following input parameters:

- Potential strength V_0
- Number of k-points (n_k)
- Number of plane waves (number of G vectors)

The program should:

(i) Construct the Hamiltonian matrix in the plane wave basis.

(ii) Diagonalize the matrix to compute the lowest five eigenvalues.

(iii) Generate data to plot the energy dispersion relation E vs k.

(iv) Generate data to plot the charge density:

- For the lowest energy eigenvalue at $k = 0$
- For the lowest two energy eigenvalues at $k = -\frac{\pi}{a}$

Plot the wavefunctions in the range:

$$-2.5a \leq x \leq 2.5a$$

Use $a = 1.0$ and $n_k = 401$.

Further, take $V_0 = 0.0$ and compute the total energies using different numbers of plane waves:

$$N = 3,\ 11,\ 21,\ 61$$

- Plot the energies for each case.
- Comment on how the results change with increasing number of plane waves.
- Can this be used to determine an optimal number of plane waves? If so, what is a reasonable choice?

Answer:

Physical Insight for $V_0 = 0$

When $V_0 = 0$, the system reduces to a free electron model. The Hamiltonian becomes diagonal:

$$H_{G,G'}(k) = \frac{1}{2}(k + G)^2 \delta_{G,G'}$$

There are no off-diagonal terms since the potential is zero. Therefore, each plane wave is already an eigenstate.

Effect of Number of Plane Waves

- Increasing the number of plane waves increases the size of the Hamiltonian matrix.
- However, since the Hamiltonian is diagonal for $V_0 = 0$, adding more plane waves simply adds higher-energy states.
- The lowest energy eigenvalues (bands) remain unchanged regardless of the number of plane waves.

Observation

We observe that for $V_0 = 0$, all calculations using $N = 3, 11, 21$, and 61 plane waves yield identical results for the lowest energy bands.

Conclusion on Convergence

- Since the results do not change with increasing number of plane waves, convergence is trivially achieved.
- This occurs because there is no coupling between plane waves in the absence of a potential.

Optimal Choice of Plane Waves

- For $V_0 = 0$, even a very small number of plane waves (e.g., $N = 3$) is sufficient.

Key Takeaway

The plane wave basis size affects accuracy only when the potential introduces coupling. In the free electron case, all basis sizes give identical physical results for the low-energy spectrum.

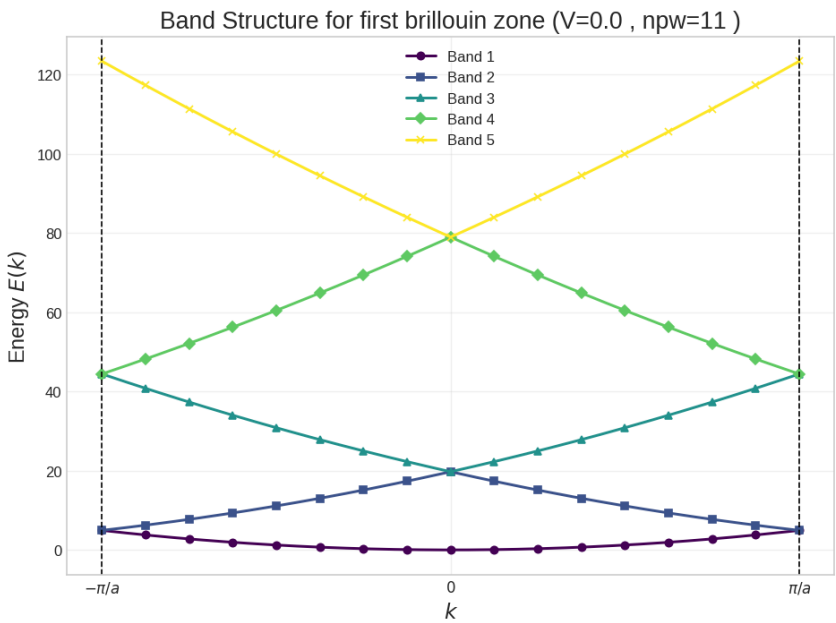


Figure 1: First brillouin zone for npw=11

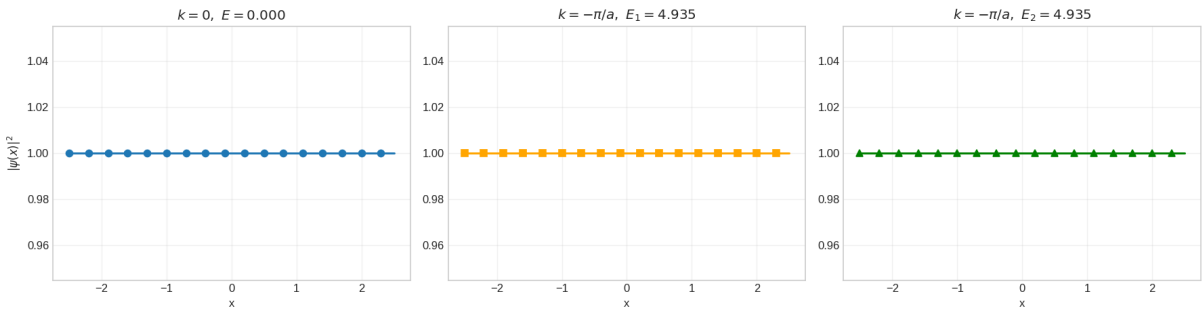


Figure 2: Electron density as func of distance for plane wave

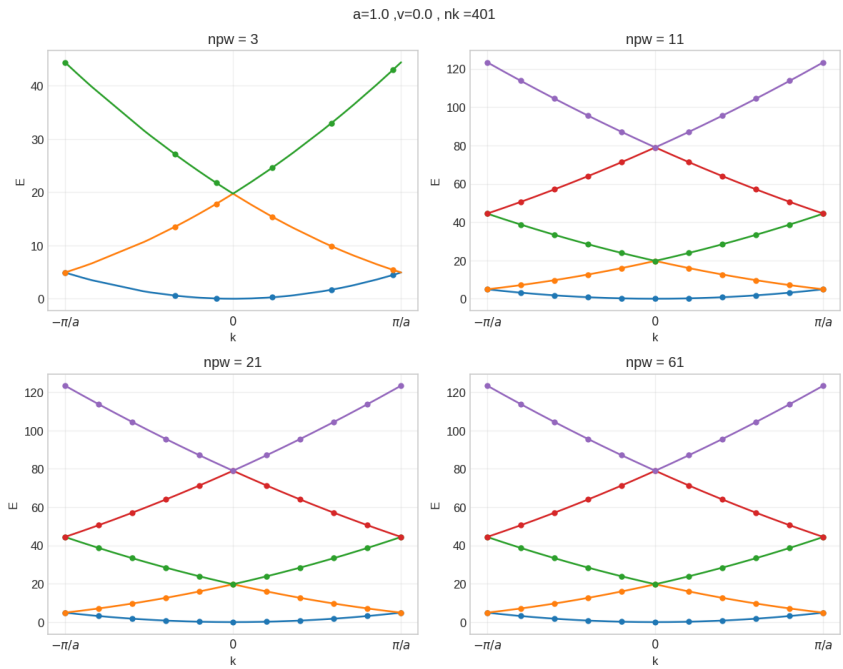


Figure 3: First brillouin zone for different npw, V=0

Question 4

For the case in part (3), take $V_0 = 0.0$ and vary the lattice constant to:

$$a = 1.5 \quad \text{and} \quad a = 2.5$$

- Compute the energy bands using different numbers of plane waves.
- Compare the results with the case $a = 1.0$.
- How do the computed quantities change with increasing a?
- Explain your observations physically.

Answer

We observe that as the lattice constant a increases from 1.0 to 1.5 and 2.5, the energy of all states decreases.

Mathematical Explanation

For $V_0 = 0$, the system reduces to the free electron model with eigenvalues:

$$E = \frac{1}{2}(k + G)^2$$

where the reciprocal lattice vectors are:

$$G = \frac{2\pi n}{a}$$

As a increases:

- The spacing between reciprocal lattice vectors decreases.
- Hence, the magnitude of G becomes smaller.

Therefore:

$$E \propto (k + G)^2 \sim \left(\frac{1}{a}\right)^2$$

which implies:

$$E \propto \frac{1}{a^2}$$

Physical Interpretation

- Increasing a means the lattice spacing becomes larger.
- This corresponds to a larger real-space "box" for the electron.
- A larger spatial extent reduces confinement, leading to lower kinetic energy.

This is analogous to the particle-in-a-box result:

$$E_n \propto \frac{1}{L^2}$$

Conclusion

- As a increases, all energy levels decrease.
- The band structure compresses toward lower energies.
- The scaling follows approximately:

$$E \sim \frac{1}{a^2}$$

Key Insight

Larger lattice spacing reduces the kinetic energy scale of the system because the allowed momentum states become more closely spaced in reciprocal space.

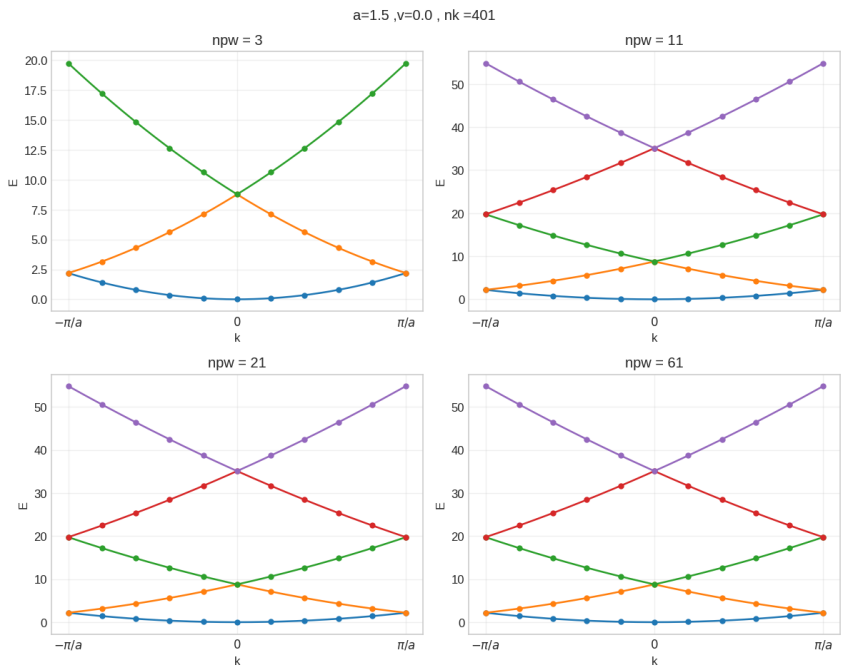


Figure 4: Brillouin zone for different npw ,a=1.5

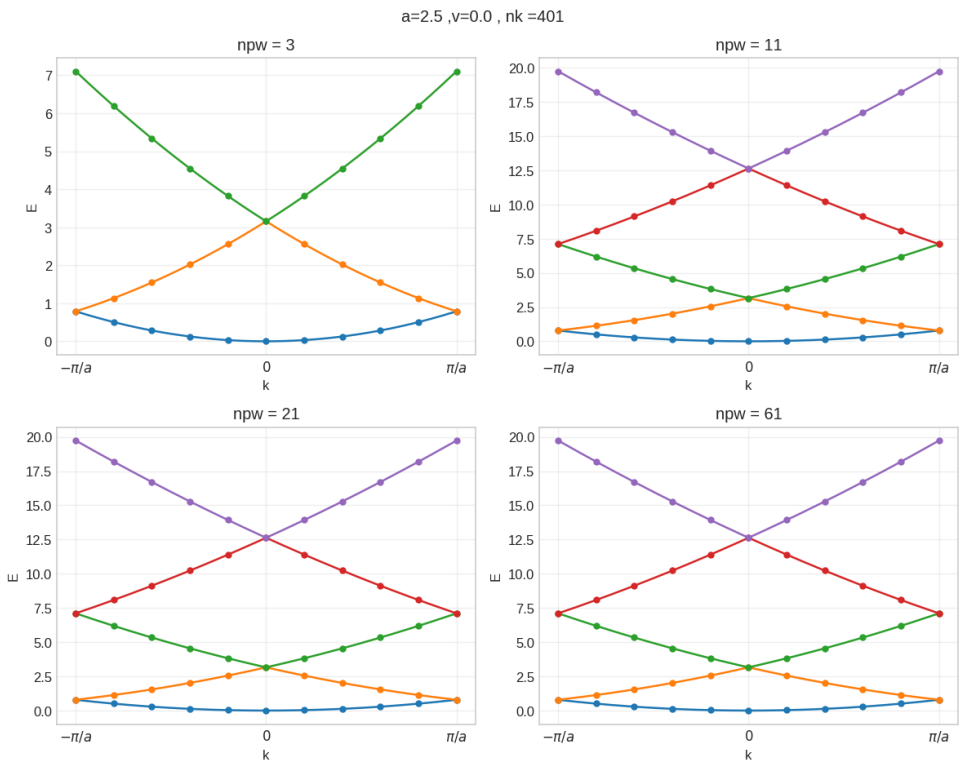


Figure 5: Brillouin zone for different npw , a=2.5

Question 5

Now consider a finite periodic potential with strength $V_0 = 0.5$.

- How many plane waves should be used for this case?
- Compute and plot the quantities mentioned in Question 2:
  1. Band structure E vs k
  2. Charge density for:
     - Ground state at $k = 0$
     - Lowest two states at $k = -\frac{\pi}{a}$
- What changes do you observe compared to the $V_0 = 0$ case?
- Are these results consistent with analytical expectations?

Answer

Choice of Number of Plane Waves

Unlike the $V_0 = 0$ case, the Hamiltonian now contains off-diagonal terms due to the periodic potential. This leads to coupling between plane waves.

- A small number of plane waves is no longer sufficient.
- Convergence must be checked numerically.
- From comparison (as in Q3), a reasonable choice is:

$$N \approx 11$$

For $V_0 = 0.5$, using $N \sim 11$ plane waves already gives well-converged lowest bands.

Observed Changes

Compared to the $V_0 = 0$ (free electron) case:

- **Band gaps open** at the Brillouin zone boundaries ($k = \pm\frac{\pi}{a}$).
- Energy dispersion deviates from the simple parabolic form.
- Degeneracies at zone boundaries are lifted.
- Charge densities become spatially modulated instead of uniform.

Physical Explanation

- The periodic potential couples states with wavevectors differing by reciprocal lattice vectors.
- At $k = \pm\frac{\pi}{a}$, strong coupling occurs between degenerate states.
- This leads to splitting of energy levels and formation of band gaps.

Charge Density Behavior

- At $k = 0$, the ground state density reflects the periodicity of the lattice.
- At $k = -\frac{\pi}{a}$, the two lowest states show distinct spatial patterns:
  - One localized near potential minima
  - One localized near potential maxima

Comparison with Analytical Expectations

These results are consistent with analytical predictions from nearly-free electron theory:

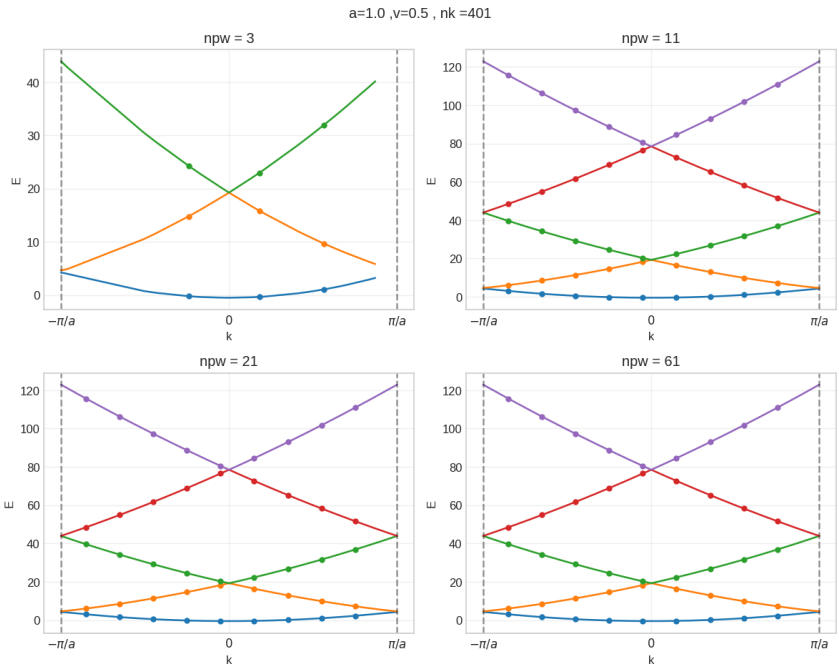
- Band gaps appear at Brillouin zone boundaries.
- Gap size is proportional to the Fourier component of the potential:

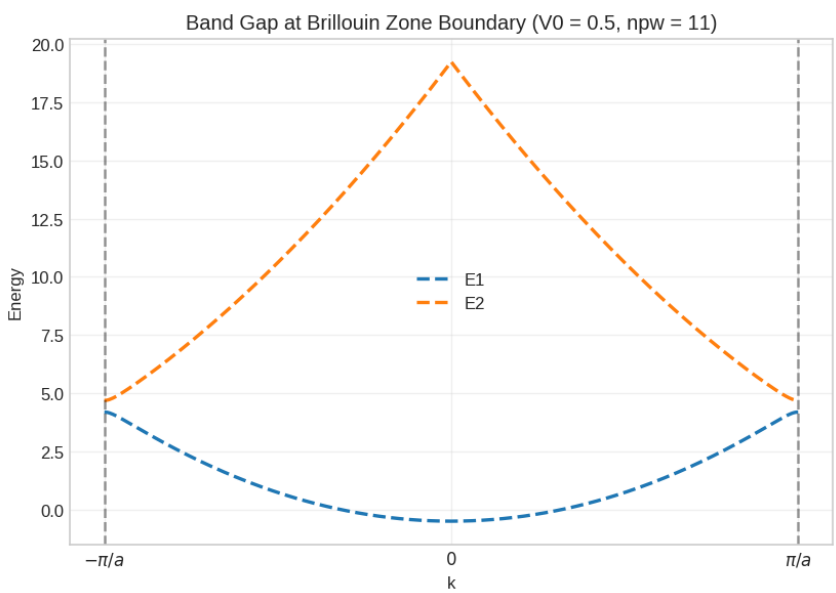
$$\Delta E \sim V_G$$

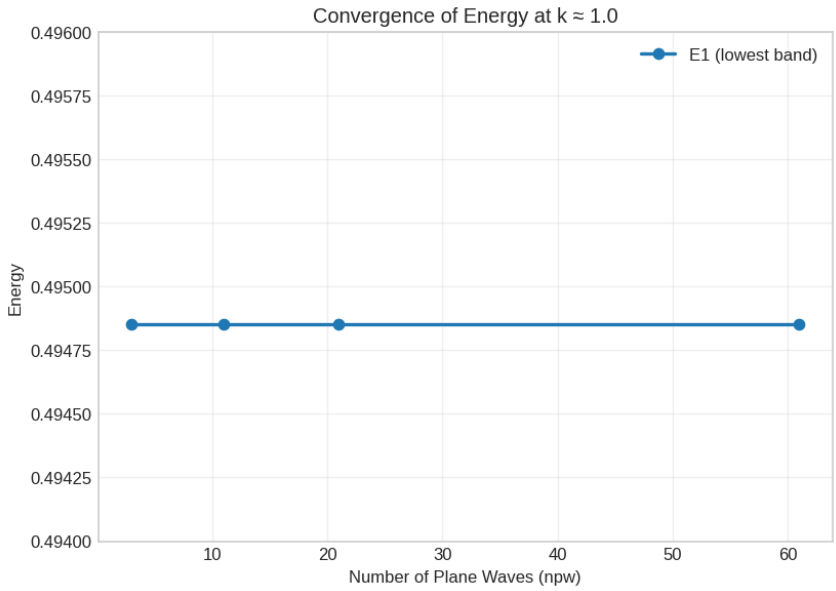
- Wavefunctions near the gap are standing waves formed by superposition of $\pm k$ states.

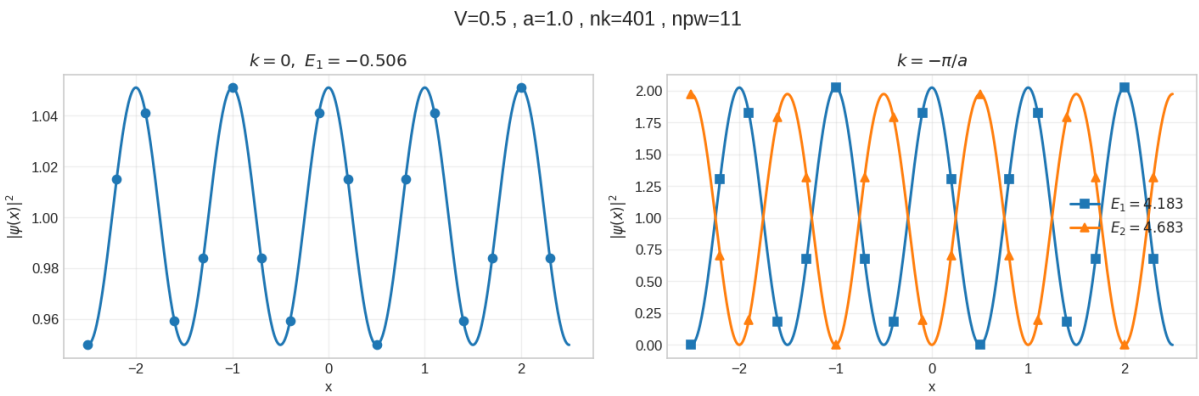
Conclusion

- Finite V_0 introduces coupling between plane waves.
- This leads to band formation and energy gaps.
- Numerical results agree well with analytical nearly-free electron theory.






Convergence of Energy at k ≈ 1.0
E1 (lowest band)
Energy
0.49600
0.49575
0.49550
0.49525
0.49500
0.49475
0.49450
0.49425
0.49400
10
20
30
40
50
60
Number of Plane Waves (npw)


V=0.5 , a=1.0 , nk=401 , npw=11
$k = 0,\ E_1 = -0.506$
$|\psi(x)|^2$
1.04
1.02
1.00
0.98
0.96
−2
−1
0
1
2
x
$k = -\pi/a$
$|\psi(x)|^2$
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$E_1 = 4.183$
$E_2 = 4.683$
−2
−1
0
1
2
x

Question 6:

At what value of V_0 does perturbation theory break down?

Answer

Perturbation theory is valid only when the perturbing potential is small compared to the characteristic energy scale of the unperturbed system.

Relevant Energy Scale

For the free electron system, the characteristic energy scale is given by the kinetic energy difference between coupled states.

At the Brillouin zone boundary ($k = \frac{\pi}{a}$), the relevant states are:

$$k = \frac{\pi}{a}, \quad k - G = -\frac{\pi}{a}$$

The corresponding energy is:

$$E_0 = \frac{1}{2}\left(\frac{\pi}{a}\right)^2$$

Criterion for Breakdown

Perturbation theory breaks down when the potential strength becomes comparable to this energy scale:

$$V_0 \sim E_0 = \frac{\pi^2}{2a^2}$$

Numerical Estimate

For $a = 1.0$:

$$E_0 = \frac{\pi^2}{2} \approx 4.93$$

Thus, perturbation theory begins to break down when:

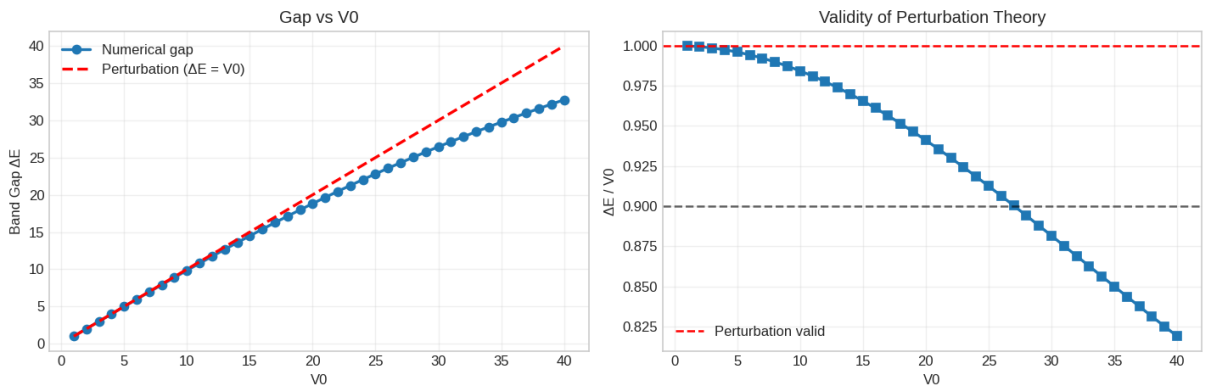
$$V_0 \sim 1 \text{ to } 5$$

Physical Interpretation

- For small V_0, the periodic potential slightly perturbs free electron states.
- As V_0 increases, strong mixing occurs between plane waves.
- Near the Brillouin zone boundary, degenerate states strongly hybridize.
- When V_0 becomes comparable to the kinetic energy scale, perturbation theory is no longer valid.

Conclusion

- Perturbation theory is valid for:
$$V_0 \ll \frac{\pi^2}{2a^2}$$
- It breaks down when:
$$V_0 \sim \mathcal{O}\left(\frac{\pi^2}{2a^2}\right)$$
- For $a = 1$, this corresponds roughly to:
$$V_0 \gtrsim 1$$



Question 7:

Now keeping the potential strength fixed to:

(i) $V_0 = 0.5$

(ii) $V_0 = 8.5$ (a.u.)

increase the lattice constant from:

$$a = 1.0 \text{ to } 10.0 \quad \text{in steps of } 3.0$$

- Compare how the computed quantities (band structure, charge density) change with respect to $a = 1.0$.
- Compare the behavior between cases (i) and (ii).
- Rationalize your observations.

Answer

General Effect of Increasing a

As the lattice constant a increases:

- Reciprocal lattice vectors decrease: $G = \frac{2\pi}{a}$
- Kinetic energy scale reduces:

$$E \sim \frac{1}{a^2}$$

- Brillouin zone shrinks in k-space

Thus, overall energy bands shift downward and become narrower.

Case (i): Weak Potential ($V_0 = 0.5$)

- System behaves close to the nearly-free electron model.
- Band structure remains approximately parabolic.
- Small band gaps appear at zone boundaries.
- As a increases:
  - Bands compress (energies decrease)
  - Band gaps become less pronounced relative to bandwidth
- Charge density shows weak modulation.

Case (ii): Strong Potential ($V_0 = 8.5$)

- System enters the tight-binding-like regime.
- Electrons become more localized in real space.
- Bands are narrow even at $a = 1.0$.
- As a increases:
  - Bands become extremely flat (very small dispersion)
  - Energy levels cluster into nearly discrete levels
  - Strong localization of charge density at lattice sites

Comparison Between (i) and (ii)

- $V_0 = 0.5$:
  - Delocalized states
  - Wide bands
  - Weak gaps
- $V_0 = 8.5$:
  - Localized states
  - Narrow bands
  - Strong gaps

As a increases:

- Weak potential case smoothly scales like free electrons.
- Strong potential case approaches isolated atom limit.

Physical Interpretation

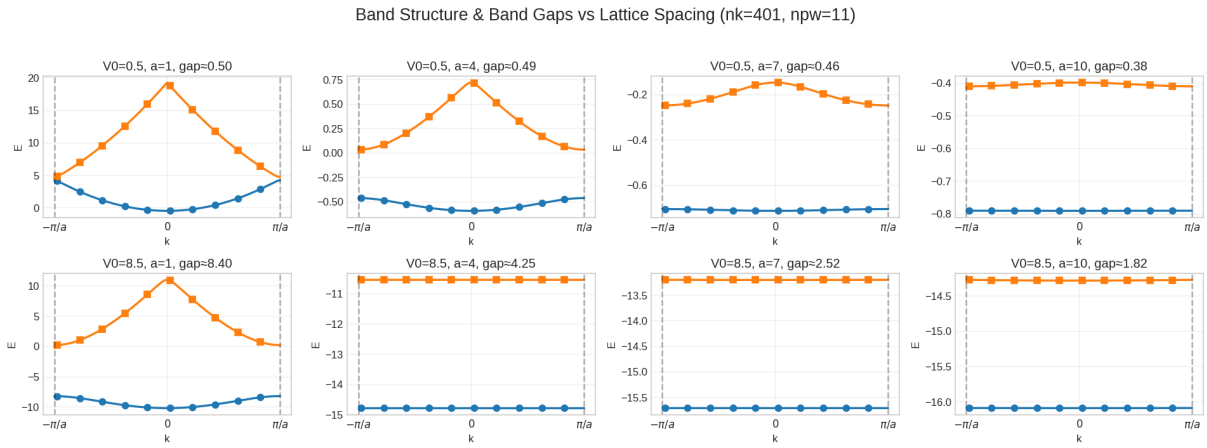
- Increasing a reduces overlap between neighboring sites.
- In weak potential:
  - Kinetic energy dominates $\rightarrow$ delocalized behavior
- In strong potential:
  - Potential dominates $\rightarrow$ localization

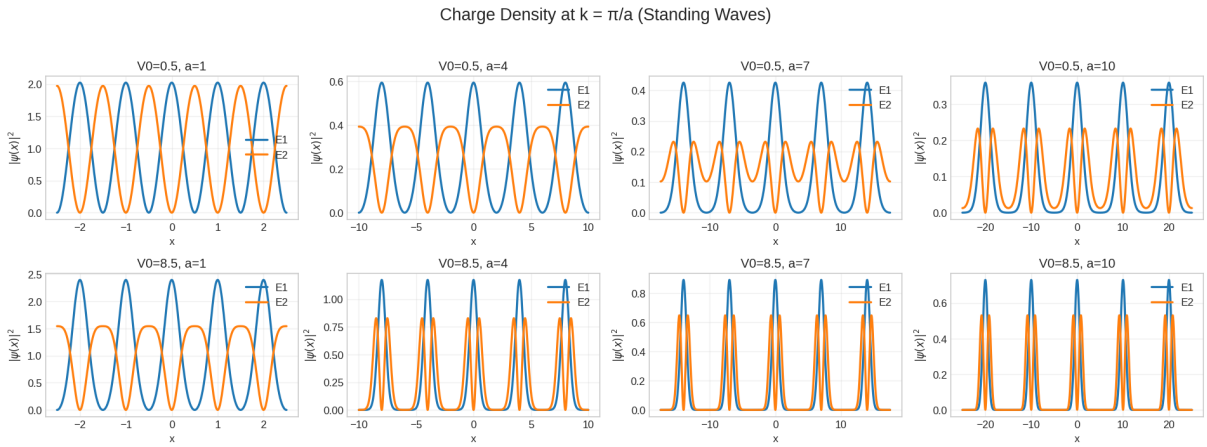
Thus, the system interpolates between:

$$\text{Nearly-Free Electron} \longrightarrow \text{Tight Binding}$$

Conclusion

- Increasing a lowers energy scales and narrows bands.
- For small V_0, system remains delocalized.
- For large V_0, system becomes localized with flat bands.
- Results are consistent with known limits of solid-state theory.


Band Structure & Band Gaps vs Lattice Spacing (nk=401, npw=11)
V0=0.5, a=1, gap≈0.50
V0=0.5, a=4, gap≈0.49
V0=0.5, a=7, gap≈0.46
V0=0.5, a=10, gap≈0.38
V0=8.5, a=1, gap≈8.40
V0=8.5, a=4, gap≈4.25
V0=8.5, a=7, gap≈2.52
V0=8.5, a=10, gap≈1.82
E
k
−π/a
0
π/a


Charge Density at k = π/a (Standing Waves)
V0=0.5, a=1
V0=0.5, a=4
V0=0.5, a=7
V0=0.5, a=10
V0=8.5, a=1
V0=8.5, a=4
V0=8.5, a=7
V0=8.5, a=10
E1
E2
|ψ(x)|²
x